Bence Hervay

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Education

Computer Science BA — University of Cambridge (Trinity College) *Oct 2021 – Jun 2024*
- Second best Computer Science Dissertation (92.5%) in the cohort of 120+ students
- Full Scholarship based on academic merit — awarded to only 5 students per year (£60k p.a.)
- 75% average across all modules — strong 2:1 overall (1st placing ray tracing project)

Advanced Mathematics Class — Fazekas Mihály Grammar School *Sep 2015 – Jul 2021*
- National Olympiads — 1st in Mathematics (2018); 1st in Physics (2020); consistently top 10 throughout high school
- FGC Robotics World Champions (2017); invited to the White House in celebration of our victory against 150+ countries
- Diverse results across STEM (25+ national top 3 placements in Mathematics, Computer Science, Physics, and Robotics)
- A*A*A* equivalent final grades; Fazekas Medallion of Excellence for outstanding performance throughout high school

Artificial Intelligence Technology Training — KÜRT Academy *Jan 2019 – Jun 2019*
- Project-based in-depth ML course; completed as the youngest participant ever (at the age of 16)

Projects & Research

Non-Rupertness of the Rhombicosidodecahedron — Rust & C++ & Python & Lean *Jun 2024 – Oct 2026*
- Proved the central open conjecture of a 333-year-old mathematical topic of which convex polyhedra fit through themselves
- Computer-assisted proof involving exact arithmetic reductions and interval arithmetic to bound floating-point rounding errors
- Developed novel solutions to numerous complex sub-problems through 1,000+ hours of independent, self-motivated research

Self-play Reinforcement Learning for 2048 — C++ (Cambridge Dissertation) *Sep 2023 – May 2024*
- Developed the world's highest-scoring algorithm and the fastest simulation engine for the 2048 Puzzle Game
- The program constructs a 2048-tile in 1.2 milliseconds with 99.98% certainty and even the 32768-tile in 75% of games
- Trained a custom 200M-parameter RAMnet baseline on 150M games without any human prior on optimal heuristics
- Automatic switching between expectimax search and locally brute-forced critical phases to maximise overall score

Selected Additional Projects *~2014 – Present*

Each built without relying on high-level libraries to ensure deep understanding of the fundamentals
- Largest Rubik's Cube ever solved — 200,000 side length with a custom developed, highly optimised Rust algorithm
- GM-level chess engine — Awarded a 1-year research scholarship by the Hungarian National Research Excellence Program
- 3D path tracer — Won the Cambridge rendering competition organised across the full cohort of 120 students
- Project Euler — Top 0.04% of all users, solved ~250 problems in total (including all of the first 222)
- Stacked Multi-Wire Proportional Chamber — Invited to the Large Hadron Collider to perform atmospheric muon tomography
- Handwritten digit classifier CNN with a GUI from scratch; Rubik's cube solver LEGO robot; platformer game based on time travel; Sophisticated algorithms for Multiplayer Pac-Man, Snake, Nonogram, Sudoku, and numerous other strategic games, often SOTA

Work Experience

Quantitative Research Engineer — AGITProp *Sep 2024 – Present*
- Automatic detection and statistical refinement of financially interpretable factors for predicting covariance matrices
- Primary ownership of numerous key modules in the pipeline (including the main statistical components)
- Key role in project planning and coordination, contributing to firm-level prioritisation decisions
- Constant expansion and maintenance of the infrastructure (e.g. quantitative modules, data pipeline, testing)

Software Developer Intern — DRW *Jun 2023 – Sep 2023*
- Created a fully automated monitoring and resource allocation tool used by 200+ employees daily
- Utilised NLP techniques to filter, classify, and distribute bug reports to maximise efficiency

Quantitative Research Intern — WorldQuant *Jun 2022 – Aug 2022*
- Generalised index rebalance alphas by automatic inference of rebalance schedules and methodologies
- Developed a framework to mass-produce low-correlation alphas using random decision forests

Senior Group Leader — Alphacademy *Aug 2021 – Jun 2023*
- Taught programming to 100+ students from beginner to Olympiad level on a weekly basis
- Member of the founding team; onboarded and managed the work of 20+ teachers

Skills

Python & C++ with 200,000+ lines of experience (pre-AI), quantitative and statistical methods, ML, DSA, low-level optimisation

Cross-disciplinary STEM expertise, analytical research mindset to tackle theoretical and practical problems from first principles

Extracurricular

Rubik's Cube speedsolving — 14-second average; 20x national top 20
Ultimate Frisbee — UK National Championship 6th (2023)
Great Court Run — 3rd (2021); 2nd (2022); 1st & Beat the Clock (2023)
Face Team Acrobatic Basketball — World Cup 4th (2018)

Activities

Always happy to discuss interesting puzzles and brain teasers
Staff & Teacher — European Summer Program on Rationality
Events Officer — Cambridge University Hungarian Society
Captain and coach — Trinity College Ultimate Frisbee Society